# QuickBite — Assessment 3 Report

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**Assessment:** SJ Innovation Assessment 3 (continuation of Assessment 2)
**Issued:** Thursday, 1 October 2026
**Report date:** Sunday, 12 October 2026 — Dev Control Tower ( `assessment-3` tag)

| Link | URL |
|---|---|
| Live app | https://quickbite-nine-phi.vercel.app/ |
| GitHub | https://github.com/sanketrotangar-sji/quickbite-food-delivery-app |
| Release tag | `assessment-3` |
| Supabase project | `motqehtswgjbbvoazarh` |

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## Assessment brief (one line)

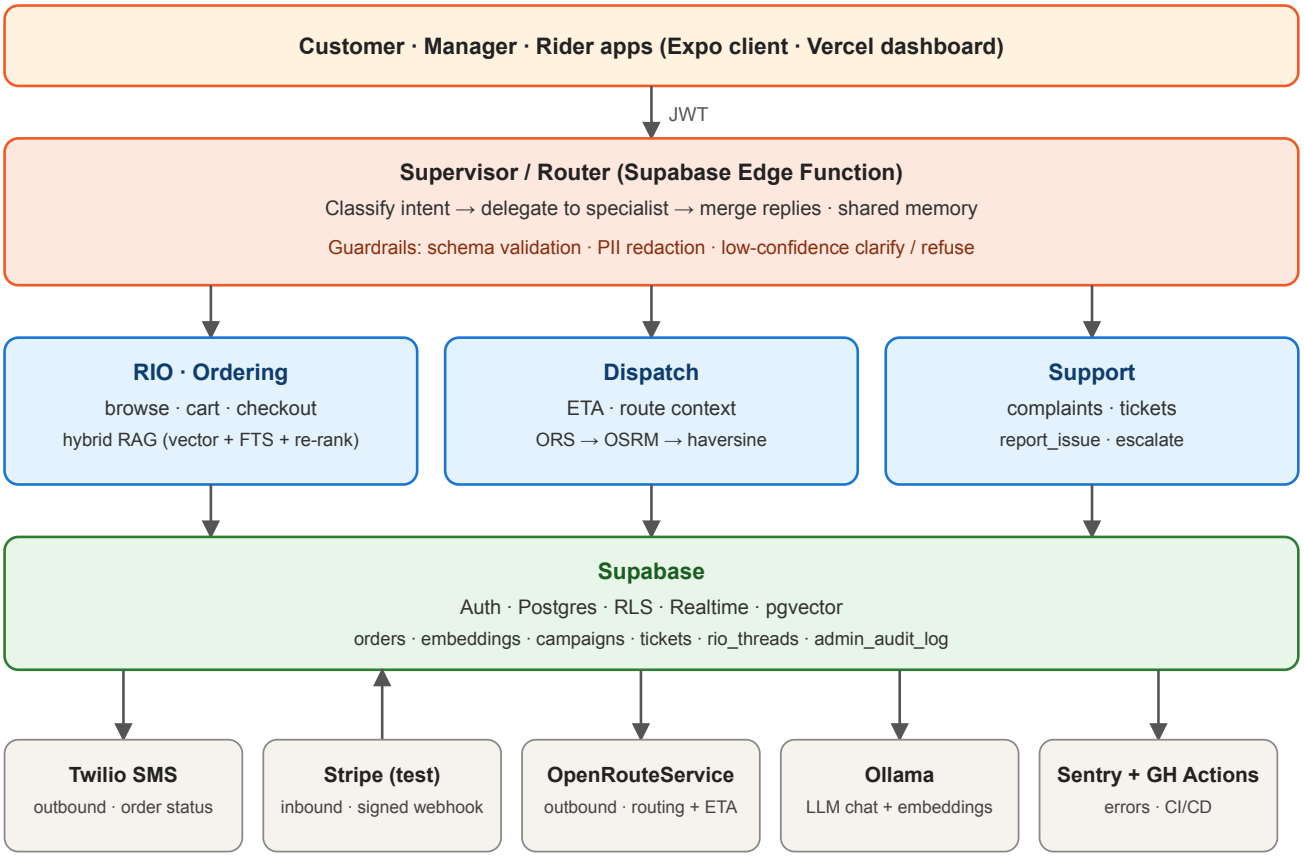
Take the Assessment-2 QuickBite app and make it **integrated** (real third-party services + campaign/CRM), **intelligent** (multi-agent with guardrails), **measured** (RAG eval harness with hybrid + re-rank), and **production-grade** (CI/CD, observability, security).

### Where Assessment 2 left off

| Cap | A2 result |
|---|---|
| RIO tool-calling agent | Cart / checkout / track / escalate |
| Automations | Smart rider-assign + kitchen-load + audit trail |
| Embeddings | Ollama (upgraded to `qwen3-embedding:0.6b`, 1024-d) |
| Menu-category classifier | **97.4%** accuracy |
| RAG matching hit-rate@5 | **37.5%** (honest — weak retrieval) |

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## Architecture (agents, integrations, data flow)




**Shared memory:** `rio_threads` / `rio_messages` + `profiles.rio_memory` + supervisor scratch.
**Guardrails:** schema validation, PII redaction, low-confidence clarify/refuse, structured JSON outputs.

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## Track 1 — Third-party integration **20 pts · complete**

≥2 real free-tier services. **One outbound + one inbound webhook with signature verification**.

| Service | Direction | Role |
|---|---|---|
| **Twilio** (SMS trial) | **Outbound** | Order-status SMS after RPC status changes ( `send-order-sms` ; Verify SMS on trial) |
| **Stripe** (test mode) | **Inbound webhook** | Verifies `Stripe-Signature` ; updates payment / refund + audit |
| **OpenRouteService / Maps** | Outbound (supporting) | Rider routing + ETA for dispatch agent |

**Outbound — Twilio:** Edge sends on placed / ready / out_for_delivery / delivered; deduped via `notification_dispatch_log` . Secrets only in Supabase Edge ( `TWILIO_*` ), never `EXPO_PUBLIC_*` .

**Inbound — Stripe:** `create-payment-intent` + `stripe-webhook` ; rejects unsigned payloads; test-mode payment/refund → order fields + `admin_audit_log` . Client optional card pay on tracking; cash default.

**Bonus:** Twilio + Stripe exercised end-to-end on the hosted project and Vercel app (test numbers / Stripe test cards).

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## Track 2 — Campaign / CRM module **15 pts · complete**

Custom loyalty / offers module in Supabase — segments, campaign outreach, redemption tracking. Admin UI: `/admin/crm`.

| Segment | Rule (summary) |
|---|---|
| **new** | ≤1 delivered order (or account age window) |
| **frequent** | ≥N delivered orders in rolling window |
| **churned** | Had orders, then idle beyond threshold |

- **Campaign builder:** title, discount %, target segment, window; Launch enrolls + in-app notify.
- **Redemption:** per customer × campaign with unique constraints; admin enrolled / redeemed / expired counts.
- Tied to `place_order` discount path so redemption is real.

## Track 3 — Multi-agent orchestration + guardrails **20 pts · complete**

| Agent | Responsibility | Example tools |
|---|---|---|
| **Supervisor / router** | Classify intent, delegate, merge replies | route → order / dispatch / support / clarify |
| **RIO (ordering)** | Discovery, cart, checkout, hybrid RAG | browse, cart, checkout, `retrieve_context` |
| **Dispatch** | ETA / route context | `estimate_eta` (ORS → OSRM → haversine) |
| **Support** | Complaints, tickets | `order_status`, `report_issue` |

Guardrails on all agents: input/output schemas, PII redaction before LLM/logs, low-confidence clarify/refuse, structured cards to the client. Shared memory across turns and specialists. Route logged on `rio_events`.

## Track 4 — RAG evaluation harness **15 pts · complete**

### What changed vs Assessment 2

| Layer | Assessment 2 | Assessment 3 |
|---|---|---|
| Retrieval | Vector-only cosine | **Hybrid:** vector + Postgres FTS / word similarity |
| Ranking | Top-k by similarity | **Re-rank** (LLM listwise + heuristic fallback) |
| Chunking / text | Thin name·category·description | Richer menu text ( `buildMenuItemText` ) |
| Corpus | Curated ~388 menu | Synthetic scale-up (**20,500** menu rows) + hybrid index |
| Labels | Fixed UUID sets | `canonical_key` / rebound dish-name gold |
| RPC | `match_embeddings` | `match_embeddings_hybrid` wired into RIO |

### RAG metrics — side-by-side (k=5, test split)

| Metric | Rebound baseline ( `old` ) | A3 ( `hybrid+rerank` ) |
|---|---|---|
| **Hit-rate@5** | **48.5%** | **81.8%** |
| **Precision@5** | 24.8% | **66.7%** |
| **MRR** | 0.347 | **0.695** |
| OOC refusal | 50.0% | **75.0%** |

**Jump:** hit-rate@5 **48.5%** → **81.8%** (above ≥80% aim). Historical A2 footnote: **37.5%**. Full write-up: `docs/rag-a3/results.md`.

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## Track 5 — Model improvement **10 pts · complete**

Order-cancellation risk (held-out test) — precision / recall / F1 (not accuracy alone).

| Metric | Value | Target |
|---|---|---|
| Precision | **0.867** | — |
| Recall | **0.825** | — |
| **F1** | **0.845** | **≥ 0.80** |
| Accuracy (secondary) | 0.877 | — |

Model: **random_forest**. Dataset: 6 real closed orders + **3000** synthetic rows. Artifacts under `docs/ml/`. Agent chat path uses Cloud Ollama via `llm_config`; embeddings stay `qwen3-embedding:0.6b` (1024-d).

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## Track 6 — CI/CD, observability & security **15 pts · complete**

- **CI/CD:** GitHub Actions on push/PR — secrets hygiene + unit + Deno edge ( `.github/workflows/ci.yml` ). Integration / Playwright when repo secrets are present.
- **Observability:** Sentry envelope ingest when DSN set; `rio_events` / `admin_audit_log`.
- **Security:** RLS + security-definer RPCs; RIO rate-limit (default 40/min); directions TTL cache; no service-role in public env vars.

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## Track 7 — Architecture docs, demo & Dev CT **5 pts · complete**

| Deliverable | Note |
|---|---|
| Architecture diagram | This report (SVG above) |
| Evaluation report | Track 4 + `docs/rag-a3/results.md` |
| Model metrics | Track 5 P/R/F1 + `docs/ml/` |

| | |
|---|---|
| CI workflow | `.github/workflows/ci.yml` — paste green Actions URL on DCT |
| Companion architecture | `docs/dct/PROJECT_ARCHITECTURE_REPORT.html` |
| Credentials | Demo / assessment role accounts ( `@quickbite.demo` ) |
| DCT tag | **assessment-3** |

## Scoring parameters checklist

| Parameter | Max | How we meet it |
|---|---|---|
| Third-party integration | 20 | Twilio SMS out + Stripe webhook in (sig verify); Maps/ORS ETA |
| Campaign / CRM | 15 | new / frequent / churned + builder + `place_order` redemption |
| Multi-agent + guardrails + memory | 20 | Supervisor → RIO / Dispatch / Support |
| RAG eval harness | 15 | Rebound **48.5%** → **81.8%** hit@5; A2 footnote 37.5% |
| Model improvement | 10 | Cancel-risk **F1 0.845**; Ollama path used by agent |
| CI/CD + observability + security | 15 | GHA unit/edge; Sentry; rate-limit / cache |
| Architecture docs, demo & Dev CT | 5 | Diagram + eval + model metrics + `assessment-3` |
| **Total** | **100** | |
| **Bonus** | +10 | Live Twilio/Stripe on hosted project; Cloud Ollama on agent path |

## Dev Control Tower checklist

1. Updated live URL + GitHub ( `assessment-3` release tag)
2. Architecture diagram (agents, integrations, data flow)
3. Evaluation report — RAG metrics vs baseline
4. Model metrics (precision / recall / F1)
5. CI workflow + observability / security note
6. Role credentials for graders

## How to reproduce key evidence

```
npm run rag:eval -- --mode hybrid+rerank
npm run ml:cancel-risk
npm run check:secrets && npm run test:unit && npm run test:edge
npm run intelligence:verify
```

**Golden rule:** depth over breadth — one real outbound + one verified inbound, one solid multi-agent flow, honest before/after RAG eval on the rebound harness (**48.5%** → **81.8%** hit@5), with A2 **37.5%** as historical footnote.